Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

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***Abstract*—AI data centers can provide demand response (DR), yet a meter decrease does not show that the credited work was committed before the event or that a local reduction created network value. This paper develops a declaration-indexed verification and settlement framework for spatially coupled AI data centers. Scheduler declarations define release, allocation runtime, requested GPUs, regional assignment, and a fixed GPU nameplate. An exact contiguous start-time witness is then carried through regional aggregation, security-constrained network valuation, and nodal settlement under one submission digest. A separate validation-fitted simplex controls total and daily conditional-value-at-risk (CVaR) false credit, while execution telemetry is opened only for locked scoring. On the locked 54-day panel, the joint risk fit obtains normalized root-mean-square error (nRMSE) 0.323, false credit 1.786 MWh/day, and the F1 (harmonic-mean) score 0.453; its held-out false-credit ratio is 0.0834 on average and 0.209 at the 90th percentile, compared with 0.0869 and 0.234 for the total-budget-only ablation. The declaration-only independent tariff replay gives $F_1 = 0.919$. Across 75,326 submissions, every runtime-complete block satisfies its declared window and GPU nameplate, with a 10^{-11}-MWh aggregate energy residual; the common witness shifts 41.1565 MWh out of the event window and is evaluated in 432 locked RTS-24 cells with all 37 finite non-islanding contingencies. The result is a reproducible physical-to-market certificate whose aggregate risk fit, execution scoring, and topology assumptions remain explicitly identified.**

Index Terms—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

I. Introduction

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch GPU work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value. Public traces already expose multi-week bursty workload arrivals and tens of thousands of scheduler jobs, while facility telemetry reports GPU energy at a different measurement boundary [1], [2].

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data. Arithmetic historical baselines remain common in market practice under published market guidance [3], despite known bias and manipulation incentives [4]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [4]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide this physical state: temporal deferral and geographical balancing reduce energy cost and peak demand [5]. Data-driven flexibility and security-constrained coordination optimize spatial–temporal dispatch [5], [6]. Balancing-market and load-shifting studies provide complementary dispatch contexts. Virtual-link clearing and space–time remuneration model geographical and temporal transfers [7], [8]. Receding-horizon frequency regulation retains multi-site state and future arrivals [9]. These methods optimize operator-controlled schedules; verification of an independently submitted DR counterfactual against an auditable workload ledger remains open. Coupled-regulation models likewise provide no provenance-bound certificate [10]. Whole-facility measurements also require a bottom-up GPU-to-facility conversion; calibration, benchmark scaling, and network placement are therefore kept as separate evidence layers.

The second gap is electrical. Geographic workload control is often valued with energy prices or gross megawatts, although nodal marginal value changes when congestion or a contingency binds. Direct-current (DC) power flow and linear distribution factors provide transparent sensitivities [11], [12]; N–1 dispatch imposes credible post-contingency limits through line-outage distribution factors (LODFs) [13]. No data-center DR verifier jointly certifies workload feasibility, signed spatial response, and security-aware network value. Alternating-current (AC) feasibility is a separate nonlinear check. Public traces support observational replay but not a causal utility event, so measured execution, counterfactual construction, and network valuation must remain separate.

This paper makes three power-system contributions. 1) It defines a typed submit–execute–settle evidence chain in which a runtime-complete job witness is formed only from scheduler declarations (release, allocation runtime, requested GPUs, and regional assignment). The same indexed witness digest is rechecked at regional aggregation and reused verbatim by the RTS-24 N–1 value and settlement replay; execution telemetry is retained for post-event scoring only. 2) It formulates a workload-state verifier with an explicit information boundary and a validation-fitted total-plus-daily-CVaR risk fit, while keeping the pointwise false-credit cap in a separately auditable payment profile. It also distinguishes the training-calibrated aggregate entitlement from a declaration-only runtime-complete entitlement and certifies every submitted job's contiguous block, regional nameplate, and deadline. 3) It develops signed space–time nodal remuneration and a payment-activation rule over calibration-validation power-conversion factors: the raw q_{99} endpoint is retained under the declared fixed/flexible payment-network scale, while the independently reconciled flexible-only capacity factor is reported as a planning diagnostic and never used to clip that endpoint. The central result is a dimension-preserving coupling invariant that carries one declared workload state into physical verification, network valuation, and settlement; the individual linear-program (LP), dispatch, and allocation primitives are standard and are not claimed as new in isolation. Because public traces do not identify utility geography, spatial conclusions are conditional on the declared feature- stratified trace scenario and the complete permutation/penetration panel. The certificate assumes trusted scheduler and telemetry provenance: a digest detects post-commitment record changes but cannot authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

II. System Model and Problem Formulation

A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals; the symbol h is reserved for a calendar day in statistical panels. The measured ledger supplies arrival energy a_{skt} for source s, class k, and release interval t. The decision $x_{skd\tau} \geq 0$ is the energy served at destination d in interval τ. Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^{t} \sum_{d \in \mathcal{D}} x_{skd\tau}. \tag{1}$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^{t} a_{sk\tau}$ and deadline D_k, release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \tag{2}$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \tag{3}$$

The state recursion equals service and terminal service equals total arrivals; $\sum_{s,k} x_{skdt} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skdt}. \tag{4}$$

For rolling window end T^+ and final event-day release T^0, the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk,\max\{T^0, T^+ - D_k\}}, \tag{5}$$

which completes every job due inside the window and every event-day arrival while retaining later real arrivals. The window starts $D_{\max} + 96$ intervals before and ends $D_{\max}$ intervals after the event; omitted earlier work has expired. Equations (2) and (3), together with (4), therefore map an aggregate, preemptive service envelope to the facility meter. This envelope is valid for checkpointable batch work but does not assert nonpreemptive placement of each GPU job; Experiment 14 separately audits contiguous MIT execution intervals, runtime, GPU count, and native power. For the counterfactual task-level check, let $j \in \mathcal{J}_{\text{sub}}$ index a submitted scheduler record, r_j denote its submitted release, and let d_j be the service-window endpoint. Slurm "timelimit" is an allocation run-time declaration; it is converted to $R_j = \lceil \text{timelimit}_j / \Delta t \rceil$ slots and combined with a precommitted queue allowance $B = 96$ slots, so $d_j = r_j + R_j + B$. The unlimited Slurm sentinel is mapped to the precommitted horizon $H_\infty = 128$ slots before adding B; finite declarations are retained exactly. The 0.001-MW-per-GPU nameplate is fixed before the split. The training-calibrated entitlement $\widehat{E}_j$ is retained only for the aggregate Exp19 calibration audit; it is not used to claim task completion. The executable witness uses the declared allocation runtime itself. Let s_j be its submit-time scenario region and g_j^{req} its requested GPU count. Define the runtime-complete entitlement $E_j^{\text{run}} = R_j g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t$. The indexed service variable $u_{j\tau}$ is the energy assigned to job j in interval τ, and the executable witness requires one contiguous fixed-rate block of exactly R_j slots. Let $\bar{p}_{\text{GPU}} = 0.001$ MW/GPU be the committed nameplate, w_{batch} the predeclared waiting-cost coefficient per MWh-slot, and r^{DR} the event service tariff. Define $c_j = g_j^{\text{req}} \bar{p}_{\text{GPU}} \Delta t$ and $\Theta_j = \{r_j, \ldots, d_j - R_j\}$. For each admissible start θ, a binary $z_{j\theta}$ selects the block and $u_\tau^{j\theta} = c_j \mathbf{1}\{\theta \leq \tau < \theta + R_j\}$. The exact start-time mixed-integer linear program (MILP) is

$$\begin{aligned}
&\sum_{\theta \in \Theta_j} z_{j\theta} = 1, \qquad z_{j\theta} \in \{0,1\}, \\
&u_{j\tau} = \sum_{\theta \in \Theta_j} u_\tau^{j\theta} z_{j\theta}, \qquad \sum_{j: s_j = d} u_{j\tau} \leq \bar{P}_d \Delta t, \\
&\min_z \sum_{j, \theta \in \Theta_j} z_{j\theta} \sum_\tau \left[w_{\text{batch}}(\tau - r_j) + r^{\text{DR}} \mathbf{1}\{\tau \in \mathcal{E}\} \right] u_\tau^{j\theta}.
\end{aligned} \tag{6}$$

This exact construction enforces runtime completion, contiguity, the GPU nameplate, and regional capacity at every slot. Experiment 27 enumerates every admissible start for all 75,326 records and stores both the earliest-start declaration baseline and the tariff response; Experiment 25 solves the same declaration model with a binding regional row. The measured $[t_j^{\text{start}}, t_j^{\text{end}}]$ interval is opened only for independent post-event scoring after the declaration witness is fixed: 71,128 execution matches never constrain z, u, or d_j. The aggregate $\widehat{E}_j$ diagnostic and the runtime-complete E_j^{run} certificate therefore have distinct, explicit semantics. For the job-to-network certificate, the indexed witness is mapped without a second optimization. To avoid overloading the deadline endpoint d_j, let $\mathcal{J}_{sk\delta}$ be the ledger partition whose source, class, and declared destination labels are s, k, δ; Experiment 27 uses the native assignment

$\delta = s_j$ and therefore has no outcome-dependent migration:

$$x^{\rm job}_{sk\delta\tau} = \sum_{j\in\mathcal{J}_{sk\delta}:\, r_j\le\tau<d_j} u_{j\tau}, \tag{7}$$

$$p^{\rm job}_{\delta\tau} = P^{\rm fix}_{\delta} + \Delta t^{-1} X^{\rm job}_{\delta\tau}, \qquad X^{\rm job}_{\delta\tau} = \sum_{s,k} x^{\rm job}_{sk\delta\tau}. \tag{8}$$

Aggregate migration remains a matched-workload control; (8) is the indexed replay profile. Experiment 27 reconstructs $X^{\rm job}$ from every stored $u_{j\tau}$, rechecks the indexed primal, and verifies equality before network valuation; no separately optimized aggregate receives a network value. Workload deferral and geographic assignment follow [5], [6], while rolling-state control follows [9]; waiting and migration costs are $w_k(\tau - t)x_{skd\tau}$ and $m_{sd}x_{skd\tau}$, with service-time substitution differing only by a fixed-arrival constant.

For provenance, the submit-time ledger and post-event execution ledger have separate canonical digests. Each submitted record is represented by the scheduler-only tuple; its Secure Hash Algorithm 256 (SHA-256) digest is defined by

$$\begin{aligned} \ell^{\rm sub}_j &= (\mathrm{id}_j, t^{\rm submit}_j, \mathrm{timelimit}_j, g^{\rm req}_j, \kappa_j, \sigma_j),\\ \mathcal{L}_{\rm sub} &= \mathrm{sort}\{\ell^{\rm sub}_j : j\in\mathcal{J}_{\rm sub}\},\\ h_{\rm sub} &= \text{SHA-256}(\mathrm{serialize}(\mathcal{L}_{\rm sub}))\,. \end{aligned} \tag{9}$$

where $g^{\rm req}_j$, κ_j, and σ_j denote the requested-GPU, class, and scheduler-state fields. The execution ledger separately stores start/end times, DCGM energy, and telemetry counts; its one-to-one join, release–execution order, and raw-to-join energy equality are checked only for post-event reconciliation. The Secure Hash Standard [14] and canonical ordering make $h_{\rm sub}$ in (9) independent of execution telemetry. Experiment 16 reconciles the measured envelope with the 118-MW nameplate fixed before the validation/test split.

The verifier assumes scheduler/DCGM services are authoritative and the digest is committed before the event decision. It cannot establish that a deadline, region, class, or energy field was truthful before signing; upstream attestation is outside this certificate.

B. Counterfactual, response, and false credit

For event set $\mathcal{E}\subset\mathcal{T}$, every p symbol denotes a regional facility-power trajectory in MW and every time sum is converted with Δt to MWh. The statistical estimate $\widetilde{p}$ is the baseline fitted from the declared information set; p^0 is the trace-anchored no-event counterfactual, $p^{1,\rm sim}$ is the workload-optimized event trajectory, and $p^{\rm obs}$ is the independently measured Massachusetts Institute of Technology (MIT) Data Center GPU Manager (DCGM) execution trace. In the implementation, the trace-anchored diagnostic is the same measured no-event tensor, so $p^{\rm obs} = p^0$ only for the observational replay; the simulated source remains $p^{1,\rm sim}$. Because no utility event label is present, $p^{\rm obs}$ is never a causal outcome or a pre-decision input. In mechanism-isolation rows $r = p^0 - p^{1,\rm sim}$ is signed, so rebound is retained. For source $z\in\{\mathrm{obs},\mathrm{sim}\}$, submitted and true credit are $\widehat{r}^{(z)} = [\widetilde{p} - p^{(z)}]_+$ and $r^{(z)} = [p^0 - p^{(z)}]_+$; source-specific false credit is

$$F^{(z)} = \Delta t \sum_{d,t\in\mathcal{E}} [\widehat{r}^{(z)}_{dt} - r^{(z)}_{dt}]_+, \qquad z\in\{\mathrm{obs},\mathrm{sim}\}. \tag{10}$$

The source-specific quantity in (10) is evaluated separately for the simulated and observed trajectories. The simulated source defines the validation epigraph; the observed source is recomputed only in the locked replay. For nonnegative true credit, $p^{(\rm sim)} + r^{(\rm sim)} = \max\{p^0, p^{(\rm sim)}\}$, an audited identity rather than a new meter trajectory. The gross forecast credit is not a transfer: before the event, the operator freezes $p^{\rm con}$, the profile actually offered for the product, and after meter closure the payable response is the pointwise intersection

$$Q^{\rm pay} = \Delta t \sum_{d,t\in\mathcal{E}} \min\bigl\{[\widetilde{p}_{dt} - p^{\rm obs}_{dt}]_+, [p^{\rm con}_{dt} - p^{\rm obs}_{dt}]_+\bigr\}\,. \tag{11}$$

Equation (11) uses only the submitted baseline, frozen contract, and closed meter. In the payment-band panel $p^{\rm con} = p^{\rm safe}$; in the decision-time panel it is the no-tariff committed-ledger profile. These are different instantiations of the frozen contract role, not interchangeable statistical estimates. The trace-anchored p^0 is an offline over/underpayment diagnostic and never enters fitting, gate decisions, workload optimization, or transfer formation. On locked event slots, $\mathrm{nRMSE} = \sqrt{N_E^{-1}\sum_n(\widetilde{p}_n - p^0_n)^2}/\max\{N_E^{-1}\sum_n p^0_n, \epsilon\}$, while bias is the signed mean baseline error. With $M = \sum_n \min\{\widehat{r}_n, r_n\}$, precision P, recall R, and F_1 are $M/\max\{\sum_n \widehat{r}_n, \epsilon\}$, $M/\max\{\sum_n r_n, \epsilon\}$, and $2PR/\max\{P+R, \epsilon\}$, where N_E is the number of region–slot samples. All sums use the same MW/MWh conversion Δt.

C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability q, and DR price $\pi^{\rm DR}$. If reference day j adds event-window service δR_j, the expected payment change is

$$\Delta\Pi = \frac{q\pi^{\rm DR}}{H}\sum_{j=1}^{H}\delta R_j - \sum_{j=1}^{H}\Delta C_j. \tag{12}$$

Let c'_j be the right derivative of the minimum operating cost with respect to an event-window minimum-service constraint on day j. Linear-program sensitivity [15] gives the following exact local condition:

$$\frac{q\pi^{\rm DR}}{H} > \min_j c'_j \iff \exists\ \text{a strictly profitable local deviation.} \tag{13}$$

At a degenerate optimum the dual at the current service level can be zero even when the right-direction marginal cost is positive. Experiment 1 therefore certifies c'_j with two independent right finite differences of 5×10^{-4} and 10^{-3} MWh and a 10^{-6} USD/MWh agreement tolerance. The historical-baseline premise is documented in [4]; (12)–(13) are this paper's workload-specific specialization.

D. Network value

Let M be the data-center-to-bus incidence matrix and $p_t = p^{\rm native}_t + Mp^{\rm dc}_t$ the nodal demand vector. For generator-to-bus map C_g, generator bounds, and power-transfer distribution factor (PTDF) H with line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_{g} \sum_{i} C_i(g_i) \\ \text{s.t. } \mathbf{1}^\top g = \mathbf{1}^\top p, \quad \underline{g} \le g \le \bar{g}, \\ -\bar{f} \le H(C_g g - p) \le \bar{f}. \end{aligned} \tag{14}$$

The incidence mapping aggregates device demand at its connection bus, and (14) is the standard lossless DC representation [11], [12]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \tag{15}$$

Positive ΔV is a credit and negative ΔV a debit; no positive-only truncation is applied.

III. Proposed Methodology

A. Model-based framework and matched-information pre-estimation

The state, power, and network equations in Section II define the feasible state before a target or payment is formed (Fig. 1). Meter history and submitted declarations enter the pre-estimator and feasible set in parallel; execution remains scoring-only. Online baselines use a lagged meter/calendar view, whereas complete-ledger comparators share the declaration view. Ridge and gradient boosting [16], [17], extremely randomized trees, synthetic control, median-loss boosting, and nonnegative donor weighting provide the baseline ledger [18], [19]. The complete-ledger quantile output receives the same exact projection with its penalty selected by contiguous folds; execution energy and completion timestamps open only at locked scoring.

B. Sparse cumulative workload-state verifier

For statistical estimate $\widetilde{p}$ and each predeclared penalty ρ_ℓ, the first-stage program solves

$$\min_{x,y,z} \mathcal{C}_{\mathrm{w}}(x) + \rho_\ell \sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt} \ge \pm(p_{dt}(x) - \widetilde{p}_{dt}), \tag{16}$$

subject to (1)–(4). The absolute-value epigraph is standard [15]; its use for workload-ledger verification is specific here. $\mathcal{C}_{\mathrm{w}}$ is workload cost, C_i is the generator-segment cost in (14), and $V(p)$ is dispatch value, so (16) changes only the objective. Six globally solved schedules $p^{(\ell)}$ share identical arrivals and constraints. A seventh matched-information feasible-quantile projection is retained as an external transfer comparator and is included in the validation-only hull. Their convex combination $\bar{p} = \sum_{\ell=1}^{7} \alpha_\ell p^{(\ell)}$, $\alpha \ge 0$, $\mathbf{1}^\top \alpha = 1$, remains feasible because the common scheduling set is convex.

The convex fit minimizes validation squared error plus the additive normalized exposure terms below, subject to total and daily-tail mechanism-isolation false-credit budgets; conditional value at risk (CVaR) measures the latter [20]. Let $\ell^\star$ be the single candidate with minimum worst contiguous-fold nRMSE and define $B_j^{\mathrm{cap}} = F_j^{(\mathrm{sim})}(\ell^\star)$, where $F_j^{(\mathrm{sim})}$ is obtained from the explicit submitted-minus-true credit definition above. Let $E_j = \max\{\sum_{n\in j} r_n^{(\mathrm{sim})}, \epsilon_E\}$ be the validation credited-energy denominator and write $\widetilde{F}_j = F_j/E_j$ and $\widetilde{B}_j^{\mathrm{cap}} = B_j^{\mathrm{cap}}/E_j$. The denominators are fixed before the locked split and the observed-meter source is audited separately. Let $m = \lceil (1-\gamma)J \rceil$ be the number of upper-tail days. For reserve $\beta \in (0, 1]$, the exact validation program is

$$\begin{aligned} \min_{\alpha,f,\nu,u} \quad & \frac{\|\bar{p} - \widetilde{p}\|_2^2}{N s_p^2} + \lambda \|\alpha\|_2^2 + \omega_T \frac{\sum_j F_j}{S_T} \\ & + \omega_C \frac{\nu + m^{-1} \sum_j u_j}{S_C}, \end{aligned} \tag{17}$$

Equation (17) fixes both normalized exposure preferences inside the convex fit; they are not post-fit tie breakers. Here $F_j = \sum_{n\in j} f_n$, $m = \lceil (1-\gamma)J \rceil$, and $s_p^2 = \max\{N^{-1}\|\widetilde{p}\|_2^2, \epsilon\}$. The exposure scales are $S_T = \max\{\sum_j B_j^{\mathrm{cap}}, \epsilon\}$ and $S_C = \max\{\mathrm{CVaR}_\gamma(\widetilde{B}_j^{\mathrm{cap}}), \epsilon\}$. The corresponding ω_T or ω_C term is omitted in a single-axis ablation; the locked joint fit fixes $\omega_T = \omega_C = 0.50$, $\gamma = 0.75$, and $\lambda = 10^{-8}$ before the locked partition. The risk-budget constraints are

$$\sum_j F_j^{(\mathrm{sim})}(\alpha) \le \beta \sum_j B_j^{\mathrm{cap}}, \tag{18}$$

$$\mathrm{CVaR}_{0.75}\Big(\widetilde{F}_j^{(\mathrm{sim})}(\alpha)\Big) \le \beta \mathrm{CVaR}_{0.75}(\widetilde{B}_j^{\mathrm{cap}}). \tag{19}$$

For implementation, the positive part is not differentiated through a sort. For each event sample n and validation day j, the program introduces $f_n \ge 0$, a CVaR threshold $\nu \ge 0$, and tail slacks $u_j \ge 0$:

$$f_n \ge p_n^{\alpha} - p_n^{(\mathrm{sim})} - r_n^{(\mathrm{sim})}, \tag{20}$$

$$F_j^{(\mathrm{sim})} = \sum_{n\in j} f_n, \tag{21}$$

$$\widetilde{F}_j - \nu - u_j \le 0, \tag{22}$$

$$\nu + \frac{1}{m} \sum_j u_j \le \beta \, \mathrm{CVaR}_{0.75}(\widetilde{B}_j^{\mathrm{cap}}). \tag{23}$$

The linear CVaR epigraph in (23) uses the same m-day upper-tail denominator as the reported empirical CVaR for every retained axis; it avoids sorting. Here $r_n^{(\mathrm{sim})} = [p_n^0 - p_n^{(\mathrm{sim})}]_+$ is the mechanism-isolation credit; the implementation checks the equivalent threshold identity. The daily ratio is reported as CVaR and the total row in MW-slot units; neither substitutes for the frozen contract profile p^{con} in (11). Together with the simplex, box bounds, and (18), the constraints are linear. Equal normalized exposure terms keep the two-axis fit identifiable; a single-axis ablation removes the corresponding term and constraint. The resulting quadratic program (QP) is solved after a HiGHS feasible point by a sparse primal–dual trust-region method. Stationarity, epigraph, simplex, and bound residuals are recomputed and stored. Four contiguous folds select β only when every training-fold contract is feasible, using nRMSE then risk ratio. Locked noninferiority is an outcome, not a hidden selection rule. A separately reported payment-contract projection targets $\bar{p}$ while imposing, for every event sample,

$$p_{dt}^{\mathrm{safe}} \le p_{dt}^{(\ell^\star)}. \tag{24}$$

The pointwise cap in (24) is the upper side of the contractual credit band; it is evaluated on every event slot before settlement and is not substituted for the scientific risk profile.

a) Two-sided credit band.: The upper envelope alone would admit a zero-credit profile, so a lower band is tied to the independently fitted convex target. Define the componentwise risk floor $p_{dt}^{\mathrm{floor}} = \min\{p_{dt}^{(\ell^\star)}, \bar{p}_{dt}\}$, where $\bar{p}$ is the validation-

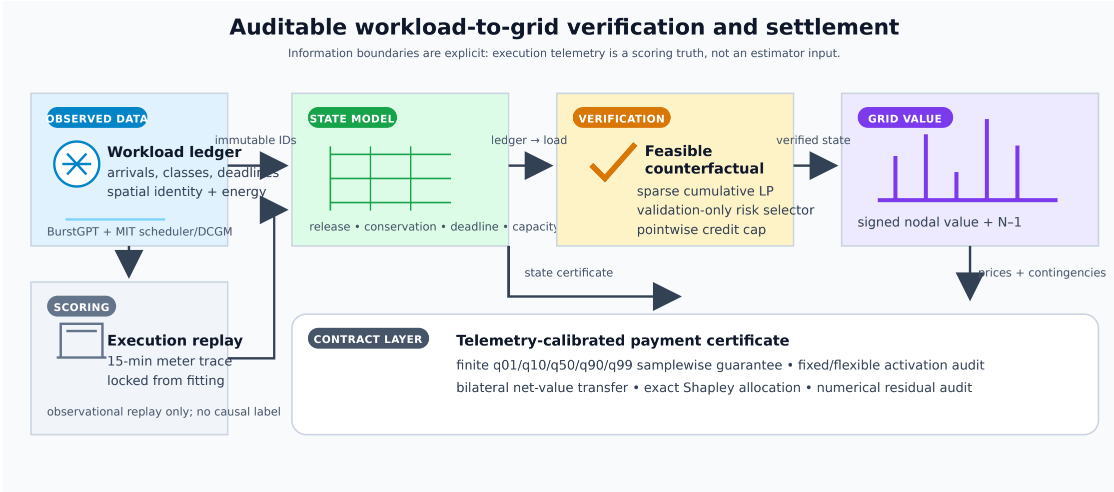

Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N–1 value and exact participant shares.

fitted convex target before the final workload projection. The event-slot constraints are

$$p_{dt}^{\text{floor}} - \varepsilon \leq p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^\star)}, \qquad (d,t) \in \mathcal{D} \times \mathcal{E}, \tag{25}$$

where $\varepsilon = 1$ MW is fixed before the locked split. The lower bound is imposed with the workload constraints, not clipped afterward, and the selected single projection proves nonemptiness. Thus the output remains attainable and can fall below the single reference when supported by the risk fit. Equations (18) and (19) control aggregate and daily-tail exposure; the upper side is the deterministic payment cap, whereas the lower side is an under-credit regularizer rather than an unseen-trace guarantee. Realized overpayment is handled by (11).

b) Decision-time information boundary.: The ex-post selector may use the complete ledger, but a gate removes all post-gate arrivals before target construction. The matched-information target is refit on this masked ledger. A no-tariff LP freezes the contract baseline and commits 5 MWh; a second LP applies the positive DR price to the same ledger and carries unexpired state to the next checkpoint. Reserved future jobs cannot be credited. Baseline error belongs to the submitted contract; the tariff plan is scored against the closed meter and settled by (11).

To separate capacity planning from payment eligibility, the gate publishes a causal reserve envelope. For candidate quantile η and event day d,

$$\widehat{a}_{sk\tau}^{\text{res}}(d;\eta) = Q_\eta(\{a_{sk\tau}(h) : h < d\}), \qquad \tau \geq t_{\text{gate}}, \tag{26}$$

Equation (26) uses earlier days only. Candidates are selected on validation days under a declared false-credit budget and used only for capacity planning; no reserved job enters payment before commitment.

The cumulative state makes storage linear rather than quadratic: explicit prefixes scale as $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|^2)$, whereas the cumulative matrix scales as $O(|\mathcal{S}||\mathcal{K}||\mathcal{D}||\mathcal{T}|)$. A 512-slot deadline is handled by (5) with every observed future arrival retained; no zero-arrival completion buffer is used in the final continuous-horizon certificate. The day trajectory is

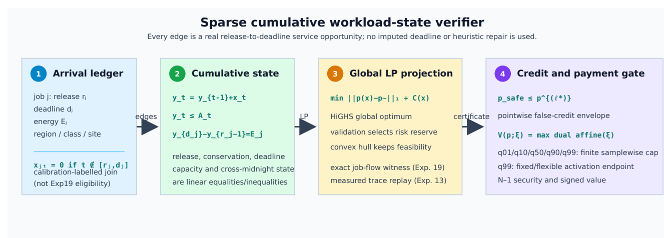

Fig. 2. Sparse cumulative verifier detail. Each admissible edge is a declared release-to-deadline service opportunity; the global projection, risk fit, and payment gate operate on the same feasible state and do not repair missing deadlines or execution records.

embedded by two lexicographic LPs: minimum L1 distance, then minimum operating cost on that face. This avoids an arbitrary penalty tradeoff; all LPs are solved to global optimality with HiGHS [21]. The resulting state-to-credit sequence is summarized by the framework in Fig. 1; the execution replay and interval audit are reported separately in Experiments 13–15. Fig. 2 resolves the ledger-to-LP path at the component level.

C. Signed nodal and N–1 settlement

Let $\lambda^0 \in \partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^\top(p^0 - p^1)$, while (15) is the realized system-value benchmark and the payoff of an explicit bilateral avoided-cost contract; it is not relabeled as an independent system operator (ISO) market payment. The convex subgradient inequality [15] yields the global certificate

$$0 \leq (\lambda^0)^\top(p^0 - p^1) - \Delta V = V(p^1) - V(p^0) - (\lambda^0)^\top(p^1 - p^0). \tag{27}$$

Thus the linear rule globally upper-bounds exact value in the implemented polyhedral market model. Equality holds only when both endpoints share a supporting dual subgradient.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from the earliest deadline-feasible anticipatory shift through the latest recovery interval. The space–time rule is

$$\Pi^{\text{ST}} = \Delta t \sum_{t \in \mathcal{W}} \sum_{d} \lambda_{dt}^0 (p_{dt}^0 - p_{dt}^1). \tag{28}$$

This is the nodal virtual-link remuneration structure of [7], [8], applied here to a verified counterfactual. Summing site payments reproduces (28) exactly.

Space–time energy remuneration is only one component of the contracted DR product. Let $\mathcal{D}^{\rm part}$ be the predeclared participating sites, $\mathcal{E}$ the event intervals, and $r^{\rm DR}$ the cleared service price. The signed delivered capacity service and total operator value are

$$Q^{\rm cap} = \Delta t \sum_{d\in\mathcal{D}^{\rm part}} \sum_{t\in\mathcal{E}} (p^{\emptyset}_{dt} - p^{1}_{dt}), \tag{29}$$

$$G = r^{\rm DR} Q^{\rm cap} + \Pi^{\rm ST}, \tag{30}$$

where $p^{\emptyset}$ is the separately re-solved minimum-cost no-event schedule on the same continuous window. It is reserved for the bilateral capacity product and need not equal the trace-anchored p^0 used in mechanism-isolation scoring. Capacity credits and participation constraints follow DR mechanism design [22], [23]; $\Pi^{\rm ST}$ debits spatial transfer and recovery. Equations (29)–(30) define signed service and operator value.

Let x^1 and $x^{\emptyset}$ denote the workload service schedules inducing p^1 and $p^{\emptyset}$, respectively. Define $C_{\rm opp} = \mathcal{C}_{\rm w}(x^1) - \mathcal{C}_{\rm w}(x^{\emptyset})$ as the participant opportunity cost and $S = G - C_{\rm opp}$ as the transferable transaction surplus. With disagreement utilities equal to zero, the symmetric Nash solution [24] activates the contract iff $S \geq 0$ and sets

$$z = \mathbb{I}\{S \geq 0\}, \quad P^{\rm B} = z\left(C_{\rm opp} + \frac{S}{2}\right), \quad u_{\rm dc} = u_{\rm op} = z\frac{S}{2}. \tag{31}$$

Thus (31) makes nonactivation the outside option: activated contracts have nonnegative utilities and the single transfer $P^{\rm B}$ is budget balanced. It is reported separately from ISO nodal remuneration and the exact dispatch-value benchmark. Unlike event-only accounting, $\mathcal{W}$ debits both pre-event load increases and delayed rebound; workload conservation gives $\sum_{d,t\in\mathcal{W}}(p^0_{dt} - p^1_{dt})\Delta t = 0$ up to solver tolerance.

Before payment comparison, the submitted, frozen-contract, and closed-meter reductions are intersected as follows:

$$q^{\rm met}_{dt} = \min\{q^{\rm sub}_{dt}, q^{\rm cap}_{dt}\}, \qquad G = r^{\rm DR}\sum_{d,t\in\mathcal{E}} q^{\rm met}_{dt} + \Pi^{\rm ST}. \tag{32}$$

The settlement chain in (32) is evaluated after the submitted profile, frozen contract, and closed meter are fixed; it is an accounting identity, not a second optimization. Experiment 12 stores the chain, site allocation, G decomposition, and bilateral budget residuals below 10^{-8} USD; these are accounting certificates rather than accuracy claims.

Security-aware settlement augments (14) with every finite non-islanding single-line outage. If $L_{\ell k}$ is the LODF from outage k to monitored line ℓ, then

$$-\bar{f}_\ell \leq (H_\ell + L_{\ell k} H_k)(C_g g - p) \leq \bar{f}_\ell, \quad \ell \neq k. \tag{33}$$

Equation (33) is the standard preventive N–1 LODF formulation [13]; all credible rows are imposed simultaneously with native ratings. Islanding outages are reported but excluded because a connected single-reference PTDF cannot represent separate islands, consistent with the declared DC/N–1 planning scope [25].

D. Scenario-robust N–1 payment certificate

The pointwise envelope controls credited energy but does not by itself certify network payment because spatially different demand vectors can activate different security constraints. Let $\mathcal{P} = \operatorname{conv}\{p^{(1)}, \ldots, p^{(L)}\}$ be the convex hull of complete workload-feasible schedules and let $p^{\rm cap} \in \mathcal{P}$ be the validation-selected single feasible projection. Its penalty $\rho_{\ell^\star}$ is selected by contiguous validation and is reported separately from the payment-target penalty. The feasible-quantile projection remains an external transfer comparator. Let Ξ_5 contain the 1st, 10th, 50th, 90th, and 99th percentiles of the calibration-validation measured-to-predicted GPU-energy ratios; these predeclared scenarios are supported by independent telemetry [2]. For facility profile $p^{\rm fac}_{dt}$, fixed demand $P^{\rm fix}_d$, and benchmark scale $s_{\rm dc}$, conversion factor ξ gives

$$p^{{\rm dc},\xi}_{dt} = s_{\rm dc}\big(P^{\rm fix}_d + \xi[p^{\rm fac}_{dt} - P^{\rm fix}_d]\big), \qquad p^{\rm fac}_{dt} \geq P^{\rm fix}_d. \tag{34}$$

The scale is calibrated on the calibration-validation q_{99} flexible peak, so the fixed component is invariant to ξ and both raw endpoints remain un-clipped activation scenarios; the flexible-only factor in Experiment 16 is diagnostic. In what follows, $Mp^{{\rm dc},\xi}_t$ is the nodal injection and superscripts α and “cap” apply (34) to p^α and $p^{\rm cap}$, respectively. For each settlement day, the payment-certified verifier defines $B^{\rm cap}_\xi = \Delta t \sum_{t\in\mathcal{E}} V_{N-1}(p^{\rm native}_t + Mp^{{\rm dc},\xi,{\rm cap}}_t)$ and solves

$$\begin{aligned}
&\operatorname{lexmin}_{\alpha,e,\eta,\{g^\xi_t\}} \quad \left(\sum_{d,t\in\mathcal{E}} e_{dt}, -\eta\right) \qquad (35)\\
&\text{s.t.} \quad p^\alpha = \sum_{\ell=1}^{L} \alpha_\ell p^{(\ell)},\\
&\qquad \alpha \geq 0, \quad \mathbf{1}^\top\alpha = 1, \quad 0 \leq \eta \leq 1,\\
&\qquad e_{dt} \geq \pm(p^\alpha_{dt} - p^{\rm safe}_{dt}),\\
&\qquad g^\xi_t \text{ satisfies (14) and (33)},\\
&\qquad\quad \text{with } p_t = p^{\rm native}_t + Mp^{{\rm dc},\xi,\alpha}_t,\\
&\qquad\qquad t\in\mathcal{E}, \quad \xi\in\Xi_5,\\
&\qquad \Delta t \sum_{t\in\mathcal{E}}\sum_i C_i(g^\xi_{it})\\
&\qquad\quad \leq (1-\eta) B^{\rm cap}_\xi, \quad \xi\in\Xi_5.
\end{aligned}$$

Generator costs use the declared piecewise-linear epigraph. Two global LPs first minimize trajectory error and then maximize the common fractional margin η on that face. Primal feasibility in (35) is equivalent to the optimal N–1 upper bound, and the unit weight on $p^{\rm cap}$ proves nonemptiness. Consequently, for any $\xi \in \Xi_5$ and realized event load $p^{1,\xi}$, daily payment obeys

$$\begin{aligned}
&\sum_{t\in\mathcal{E}} \Delta t[V_{N-1}(p^{\rm native}_t + Mp^{{\rm dc},\xi,\alpha}_t) - V_{N-1}(p^{1,\xi}_t)]\\
&\leq \sum_{t\in\mathcal{E}} \Delta t[V_{N-1}(p^{\rm native}_t + Mp^{{\rm dc},\xi,{\rm cap}}_t) - V_{N-1}(p^{1,\xi}_t)]. \qquad (36)
\end{aligned}$$

The common realized-cost term cancels, so the certificate is samplewise and needs no execution truth. Both programs contain every finite non-islanding contingency and use standard SCED/LP primitives [12], [15]. Here $\mathcal{P}$ contains the six independently solved first-stage projections from (16) and the matched feasible-quantile projection. The reported risk-constrained verifier is the validation-fitted simplex point under the total and CVaR budgets. The separately stored payment-contract profile targets it under the pointwise cap and is not fed back into the risk fit; the quantile schedule remains an independent transfer comparator.

The finite cap is complemented by a set-valued certificate. Let $p^{\mathrm{cap}} = p^{(\ell^\star)}$ be the contiguous-validation nRMSE choice, p^{pay} the separately selected validation payment target, and $p^{\mathrm{cert}} = p^{\alpha}$ the solution of (35). For realized value, define $\Delta V_\xi(p) = V_{N-1}(p,\xi) - V_{N-1}(p^{1,\xi})$ and $p_\theta = (1-\theta)p^{\mathrm{cap}} + \theta p^{\mathrm{cert}}$, $0 \le \theta \le 1$. The induced interval is

$$\mathcal{I}_\xi = \left[\min_{0\le\theta\le1} \Delta V_\xi(p_\theta),\ \max\{\Delta V_\xi(p^{\mathrm{cap}}), \Delta V_\xi(p^{\mathrm{cert}})\}\right]. \tag{37}$$

The lower endpoint is one exact joint N–1 SCED LP with a shared segment parameter; convexity gives the endpoint maximum. The cap payment belongs to $\mathcal{I}_\xi$. Endpoints use validation-frozen profiles and the realized counterfactual only; no oracle baseline selects the interval. Its width is reported as uncertainty cost, while the seven-profile hull remains the finite-scenario certificate in (35).

E. Participant allocation

For participants N, define $v(S)$ as signed dispatch value when coalition S moves from baseline to actual demand and nonmembers remain at baseline. The exact allocation is

$$\phi_i(v) = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!}[v(S\cup\{i\}) - v(S)]. \tag{38}$$

This is the Shapley value [26]; the characteristic function based on signed grid value is the proposed specialization. Grouping (38) by coalition size cancels every intermediate value and gives $\sum_i \phi_i = v(N)$ when $v(\varnothing) = 0$. All 16 coalitions for four sites and all 256 coalitions for eight contractual portfolios are enumerated; no permutation estimator is used. For larger auditable contracts, exchangeable slices at the same electrical site are grouped by their member count. The exact sum retains the binomial multiplicity of every labeled coalition and evaluates only $\prod_g(m_g+1)$ distinct count states. This symmetry reduction remains the Shapley value and is evaluated through 20 participants.

F. Analytical guarantees

These certificates delimit the logical scope. Under the interior and differentiability assumptions of (12), a feasible reference-day perturbation is strictly profitable iff $q\pi^{\mathrm{DR}}/H > \min_j c'_j$ in (13); Experiment 1 obtains the dual marginals and checks all 56 strategic cells. The workload set is linear and convex. Every simplex schedule and the two-sided envelope in (25) are feasible, and for any execution c,

$$[p^{\mathrm{safe}}_{dt} - c_{dt}]_+ \le [p^{(\ell^\star)}_{dt} - c_{dt}]_+,$$
$$[c_{dt} - p^{\mathrm{safe}}_{dt}]_+ \le [c_{dt} - p^{\mathrm{floor}}_{dt}]_+ + \varepsilon.$$

Summation gives the false-credit and under-credit bounds without locked outcomes.

The base-case and stacked N–1 dispatch values are convex and piecewise affine in nodal demand. The subgradient inequality is (27), with $0 \le \Delta V_{\mathrm{lin}} - \Delta V \le \|\lambda^1 - \lambda^0\|_* \|p^1 - p^0\|$; positive-only settlement has no such identity. In (35), each embedded dispatch is a feasible witness, so subtracting the common realized cost gives (36) for every $\xi \in \Xi_5$. The dual representation also makes $V_{N-1}(p;\xi)$ convex in a continuous conversion factor, so the worst case on $[\xi_{\mathrm{L}}, \xi_{\mathrm{U}}]$ is attained at an endpoint. The lower endpoint of (37) is the exact joint-LP minimum over the declared affine workload segment and convexity makes its endpoint maximum exact; Experiments 9 and 15 use independent 40- and four-segment models.

Finally, SHA-256 collision resistance makes post-commit changes to the scheduler-only tuple detectable in h_{sub}, while execution joins check ordering and energy conservation after the event. An optimum of (6) gives each job its entitlement within its submitted window, enforces site capacity and the GPU nameplate, and avoids aggregate rounding; contiguous execution is audited separately. With $B_{dj} = \mathbf{1}\{s_j = d\}$ and bus map M, the job, aggregation, and mapping residuals are

$$r^{\mathrm{job}}_j = \sum_{\tau=r_j}^{d_j-1} u_{j\tau} - \widehat{E}_j,$$
$$r^{\mathrm{agg}}_{d\tau} = X_{d\tau} - \sum_j B_{dj} u_{j\tau},$$
$$r^{\mathrm{map}}_{b\tau} = p_{b\tau} - P^{\mathrm{fix}}_{b\tau} - \Delta t^{-1} \sum_d M_{bd} X_{d\tau}.$$

If their typed norms are at most ε, the dispatched profile is the same indexed witness up to that tolerance; at zero residual an independently optimized aggregate cannot receive settlement value. Experiment 22 checks these residuals before N–1 valuation and preserves $\sum_i \phi_i = v(N)$.

IV. Experimental Design

A. Data, preprocessing, and information boundary

BurstGPT supplies 5,188,507 successful requests and token service [1]; MIT Supercloud supplies scheduler submissions and independent Data Center GPU Manager (DCGM) energy. The full population contains 75,326 valid submissions and 71,128 positive-energy execution matches, of which 68,664 start in the common 121-day tensor used by Experiments 1–13 [2]. Submission time defines arrivals and measured intervals define scoring truth. Chronological workload scaling uses the first 40 complete days. The GPU-power model uses 44,391 training jobs, 9,637 calibration-validation jobs, and 41,154 locked jobs; the calibration-validation ratios define the five scenarios Ξ_5 in (35) without synthetic replacement. Experiment 14 retains all 71,128 matches, whereas Experiment 19 uses every valid declaration with a training-only aggregate entitlement; Experiment 27 forms the runtime-complete declaration witness, while execution joins open only for post-event scoring. The runtime window plus a precommitted 96-slot queue allowance defines the contract window, not a submit-to-completion interpretation of Slurm runtime, and membership is never outcome-filtered.

Because the traces lack utility bus locations, four-region coupling is a trace-driven benchmark: a predeclared feature-stratified round robin assigns class, GPU count, runtime, and submission time. Experiment 11 exhausts all 24 permutations, three penetrations, and two concentration controls;

Experiment 18 uses fixed generator-bus strata at native-load multiplier 0.90, so its transfer result is conditional on that scenario. Calibration-validation ratios span 0.5827–5.3209 and the locked partition is a generalization audit. Equation (34) carries the fixed 6-MW/site component separately and calibrates payment-network scale on q_{99}; both raw endpoints remain activation-eligible. The separate flexible-only 0.8901 capacity-safe factor is planning-only, not an endpoint clip. The Power Grid Library (PGLib) and MATPOWER/PYPOWER provide topology, ratings, and costs [12], [27]; RTS-24 follows the published case [28].

B. Baselines, parameters, and inference

Experiment 2 compares meter/statistical baselines with complete-ledger quantile/feasible projections, synthetic control, tail-risk counterfactual, the selected single projection, and risk-constrained verifier; payment-certified verifier is evaluated separately in Experiments 9 and 15. Feature equality is audited while execution truth is withheld. Deferral windows are 0, 4, and 512 slots; each region has a 118-MW flexible nameplate and 6-MW fixed component. Six projection penalties and three risk reserves are selected by nested contiguous folds; normalized total/CVaR preferences and the five-point stress grid are validation-only. The gate commits 5 MWh and payment uses an independent 40-segment evaluator. Moving three-day bootstrap replicates and all 2^{18} block-sign assignments use Holm correction [29], [30]. Sparse-system size grows from 5,760 variables/13,424 nonzeros at 96 slots to 36,480/85,488 at 608 slots (0.039–0.212 s).

The model-out check uses polar alternating-current optimal power flow (AC OPF) with active/reactive balance, voltage/reactive-generation bounds, and apparent-power ratings [12]. After the intact solve, each credible outage fixes every non-reference active output at its intact value, $P_{Gi}^{(k)} = P_{Gi}^{(0)}$, while the reference active balance, reactive outputs, and voltages remain recourse. Apparent-power, voltage, and fixed-plan deviations are checked after every solve; non-reference bounds are the intact output $\pm 10^{-8}$ MW with a 10^{-6}-MW margin, and the largest deviation is 2.52×10^{-8} MW.

Experiments 1–2 test incentives/workload verification; 3–9 test signed settlement, transfer, risk, and N–1 value; 10–12 test AC/spatial transfer and continuous-cycle remuneration; 13–16 provide measured replay, indexed witness, conversion intervals, and reconciliation. Experiments 17–25 freeze the gate, verify shared-plan AC feasibility, audit scale/coupling, replay an independent event, and enumerate all RTS-24 outages. Experiment 27 adds the runtime-complete common witness, and Experiment 28 audits the held-out risk tail with a fixed block bootstrap. Comparisons use the same 54 locked days unless a deterministic network or ledger rule is stated. Experiment 26 independently recomputes the declaration-to-network residuals and checks the common witness digest, source hashes, and role labels of the risk, payment, and outage artifacts.

V. Results

A. Incentive boundary and independent verification

Experiment 1 solves the entire 8×7 DR-price/call-probability grid. The dual threshold (13) agrees with the independently solved strategic program in all 56 cells. At \$150/MWh and $q = 0.35$, strategic reference scheduling inflates the event baseline by 8.223 MWh. Of 55.452 MWh credited, 44.458 MWh is unsupported by physical reduction, giving a false-credit ratio of 0.802. The profitable region follows the derived marginal boundary and does not use a fitted classifier; the complete grid and its phase diagram are retained in the experiment output.

TABLE I
LOCKED 54-DAY MECHANISM-ISOLATION RESPONSE RESULTS (DAILY MEANS).

Method	nRMSE	False MWh	Under-credit	F_1
High-5-of-10	4.196	172.455	6.474	0.143
Ridge	2.761	109.817	5.306	0.233
Gradient Boosting	2.343	96.077	4.817	0.245
Extra Trees	3.350	133.961	6.804	0.171
Metadata Gradient Boosting	2.201	92.874	4.193	0.259
Ex-post Metadata Gradient Boosting	2.126	90.807	4.184	0.265
Ex-post Quantile Gradient Boosting	0.932	23.144	6.085	0.533
Synthetic Control	2.598	104.998	7.877	0.202
Feasible Quantile Projection	0.301	2.811	10.772	0.519
Single Feasible Projection	0.339	1.823	13.537	0.418
Tail-Risk Feasible Counterfactual	0.322	1.791	12.878	0.455
Risk-Constrained Convex Verifier	**0.323**	**1.786**	**12.906**	**0.453**

Experiment 2 evaluates twelve methods on the locked 54-day panel (Table I). The joint total-plus-daily-CVaR fit gives nRMSE/F_1/false credit 0.323/0.453/1.786 MWh/day. Its held-out daily false-credit ratio has mean 0.0834, $\text{CVaR}_{0.75}$ 0.2409, and 90th percentile 0.2090; the corresponding total-budget-only values are 0.0869, 0.2458, and 0.2340 (Fig. 3). A circular three-day block bootstrap gives a paired joint-minus-total mean-ratio difference of -0.00358 with 95% interval $[-0.00612, -0.00112]$. The total-only fit has nRMSE 0.314, whereas the joint fit spends part of that point-estimation margin to control the declared tail exposure. The total row retains 5.55 MW-slot slack and the daily-CVaR row remains feasible with normalized slack 0.00671; both budgets are fixed before the locked split. Single projection gives 0.339/0.418/1.823, whereas the unconstrained fit gives 0.297/0.521/3.241. The payment-contract cap is separate from this risk-fit objective.

The independent trace-meter replay is a deployment check: event-window MAE is 2.673/2.865/2.871 MW for feasible quantile/tail-risk counterfactual/risk-constrained verifieron the same 54 days (three-day block intervals in Experiment 20). It measures closed-DCGM alignment separately from mechanism-isolation scoring.

Closest domain baselines use identical arrivals, deadlines, capacities, event slots, and locked days. Event-reward and batch-flexibility translations give nRMSE/false-credit/F_1 3.068/114.776/0.137 and 0.339/1.823/0.418. Rolling- horizon, cross-regional dispatchable-capacity, and coupled multi-service translations use the same ledger; their equations and information sets are in the baseline audit [5], [10]. They are published-equation translations, not reproductions of private software or data; temporal-only and joint-ledger controls migrate 0 and 0.206 MWh. The ablation panel contains 810 globally solved schedules (binding shares 0.310/0.948; slope 1.003).

B. Nodal value and spatial response

Experiment 3 crosses thirteen counterfactuals with five settlement mechanisms. Table II isolates the mechanism with the trace-anchored reference and reports the locked workload

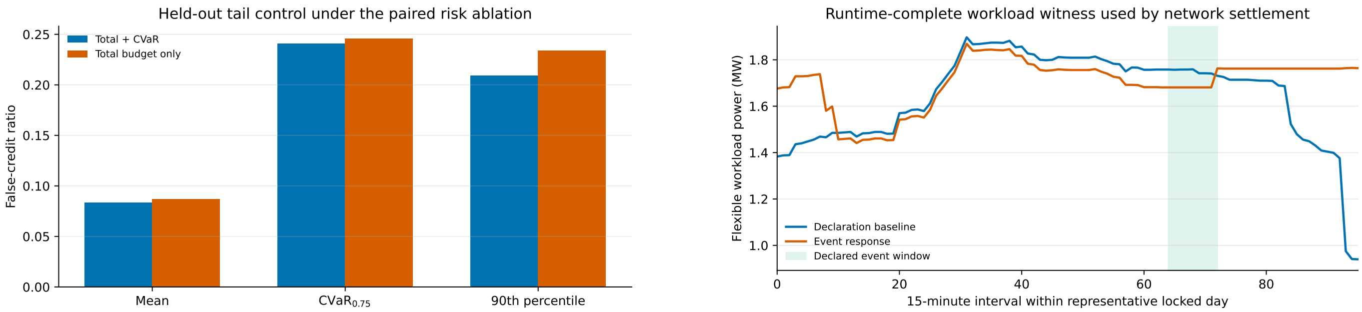

Fig. 3. Result evidence for the two coupled certificates. Left: held-out false-credit ratios for the joint total-plus-CVaR fit and the total-budget-only ablation, including the empirical upper tail. Right: declaration baseline and event response generated by the same runtime-complete job witness on a locked day; the shaded interval is the predeclared event window.

TABLE II
BASE-CASE AND COMPLETE N–1 SETTLEMENT ERRORS (USD/DAY).

Panel	Settlement	Trace MAE	Workload MAE	Trace median/overpay	Workload median/overpay
Base case	Nodal gross	360.580	421.677	–	–
Base case	Nodal exact net value	**151.719**	455.865	–	–
N–1	Base exact	139.221	167.367	79.79/130.42	121.01/107.69
N–1	N–1 linear	2.208	**150.461**	0.60/1.91	113.43/20.38
N–1	**N–1 exact**	**0.990**	150.955	0.87/0.01	113.43/20.18

rows. Exact signed nodal value has mean absolute error (MAE) \$151.719 versus \$360.580 for nodal gross response; the corresponding workload errors are \$455.865 and \$421.677. Signed netting reduces the trace error by \$208.190/day (Holm-adjusted $p = 3.89 \times 10^{-4}$), whereas uniform-to-nodal pricing changes it by $-\$1.463$ ($p = 0.0118$) and linear-to- exact value by \$0.671 ($p = 0.8026$). Workload effects are $-\$34.165$, $-\$1.191$, and $-\$0.023/day$, quantifying the mechanism-specific trade-offs on the same locked workload panel. All 5,616 interval/counterfactual instances satisfy (27); the largest violation is 2.03×10^{-11} dollars.

Experiment 4 applies the predeclared largest gross-minus-net offset (day 87), exposing the rebound component that positive-only local accounting omits.

C. Cross-network and N–1 security evidence

Experiment 5 evaluates 6,480 four-network outcomes; exact value improves the trace-anchored error in all 12 cells (mean \$1.913/day; eight Holm-adjusted tests below 0.05), with a \$0.050/day workload change. Experiment 6 solves 810 capacity/deadline schedules (binding shares 0.031–0.911; false credit 2.374–2.740 MWh).

Experiment 7 enumerates 16 four-site and 256 eight-site coalitions (budget residual 5.68×10^{-14} dollars) and 1,296 exact 20-participant states. Experiment 8 enforces (33) for all 37 RTS-24 non-islanding outages: trace/workload MAE is \$139.221/\$2.208/\$0.990 and \$167.367/\$150.461/\$150.955 for base exact/N–1 linear/N–1 exact, with 1.000 p.u. maximum loading.

Table III summarizes the complete replay scope.

Experiment 9 tests (36) on 54 global programs with a validation-frozen seven-profile hull; contiguous-fold nRMSE selects the cap and validation N–1 MAE the payment target. Factors are $q_{01} = 0.583$, $q_{10} = 0.687$, $q_{50} = 0.970$, $q_{90} = 1.349$, and $q_{99} = 5.321$, with every certificate violation below 9×10^{-9} USD. For PaymentSafe/single/feasible-quantile/RiskSafe, q50 MAE is 39.288, 32.880, 34.391, and 33.084 USD/day, with overpayment 1.642, 4.313, 3.884, and 4.241 USD/day; q99 MAE is 217.949, 183.077, 191.684, and 184.434 USD/day, with overpayment 9.472, 24.749, 22.293, and 24.388 USD/day. Moving-block intervals are stored for each comparator. Experiment 15 retains both raw endpoints without clipping; 0.8901 is planning-only. Unseen 0.80/1.20 MAE is \$8.333/\$12.502 (payment) and \$8.815/\$13.224 (feasible quantile); neither factor is certified.

Experiment 10 replaces the linear network with AC OPF [12]: all 864 connected-base and 5,400 corrective IEEE-9/14 cells are feasible, with zero deviation in 3,888 shared-plan cells (0.880/0.907/0.935 p.u. at 3/6/9%). Experiment 18 extends the fixed-active plan to four networks and 9,648 native-admissible contingency solves plus 144 intact reference solves over six snapshots. Native-inadmissible connected outages are screened before workload injection and reported by network; all remaining cells have zero voltage violation and maximum loading 1.000 p.u. Experiment 11 averages all 26 assignments across three penetrations, four counterfactuals, and 54 days; its 3/6/9% MAE is \$101.155/\$203.361/\$303.195 for RiskSafe, \$105.250/\$211.555/\$315.490 for single, and \$81.613/\$164.249/\$244.481 for feasible quantile. Experiment 16 audits 44,391/9,637/41,154 training, calibration-validation, and locked GPU records (mean absolute error (MAE), root-mean-square error (RMSE), and R^2 are 10.803/14.463/0.8956 and 13.543/20.075/0.8196 W/W/1). Experiment 12 reports event-window MAE \$26.937/\$34.166/\$8.659/\$5.325 and complete-cycle residual \$0.621/\$0.569/\$0.104/\$0 for feasible-quantile, payment-certified, RiskSafe, and single. These residuals are settlement-value accounting diagnostics, whereas event-window error is predictive; all 54 contracts activate.

D. Independent trace and deployment-boundary audits

The immutable scheduler/DCGM replay gives slot MAE 3.740 versus 5.678 MW at 100% coverage and a 71,128-job residual below 10^{-17} MWh; no utility-event label exists. Experiment 20 gives trace-meter MAE 2.988 MW (2.673 for feasible quantile). Experiment 23 instead re-enumerates every declaration-feasible start under a distinct, predeclared event

TABLE III
DECLARED INFORMATION-BOUNDARY AND SECURITY-REPLAY COVERAGE. THE INDEPENDENT DECLARATION CHECKS ARE COMPUTED BEFORE LOCKED SCORING.

Panel	Cells and certificate
Independent declaration replay	6/6 information checks; nRMSE 0.0292, $F_1 = 0.919$
Frozen-profile DC N–1 (RTS-24)	864 cells; 37 outages/cell; 1.0000 p.u.
Shared-plan AC (IEEE-9)	3,888; 0.880/0.907/0.935 p.u.
Corrective AC (IEEE-9/14)	864 base + 5,400 outage solves; all feasible
Admissible fixed-plan AC N–1 (four networks)	9,648 contingency + 144 intact solves; all certified

TABLE IV
EXACT INDEXED JOB-LEVEL AND BINDING-CAPACITY CERTIFICATES.

Quantity	Value
Submitted / execution-matched jobs; admissible starts	75,326 / 71,128; 13,198,247
Runtime entitlement; common-witness event reduction (MWh)	1,972.115; **41.1565**
Baseline / response event energy (MWh)	165.436 / 124.280
Maximum aggregate energy residual; minimum site slack (MWh)	7.8×10^{-12}; 28.595
Binding stress cohort / capacity / service floor (jobs/MW/MWh)	79 / 0.00016037 / 0.00108
Job-to-network residual; common settlement cells (MWh / count)	0; 432
DC N–1 profile panel; RTS-24 outages / cells / evaluations	37 / 864 / 31,968
Maximum counterfactual post-contingency loading (p.u.)	1.0000

tariff. Against that independent exact declaration replay, the common runtime-complete witness obtains nRMSE 0.0292, false response 0.0104 MWh/day, and $F_1 = 0.919$ over all 54 locked days. The protocol certificate verifies zero use of future arrivals or execution telemetry and an identical submission digest. This is an optimization-policy replay rather than a field-causal intervention; Table III records the information boundary explicitly.

E. Indexed workload counterfactual

Experiment 27 assigns each of 75,326 valid submissions a runtime-complete contiguous block at the declared GPU nameplate. The declaration windows and 118-MW regional capacity yield 13,198,247 admissible start choices. The runtime entitlement totals 1,972.115 MWh, the aggregate energy residual is below 10^{-11} MWh, and minimum site slack is 28.595 MWh (Table IV). The same saved baseline and event-response arrays produce 41.1565 MWh of event-window reduction and are replayed in 432 locked RTS-24 cells, each with all 37 finite non-islanding contingencies. Experiment 25 validates the separate 71,128/75,326 execution join; its 79-job declaration cohort binds 0.00016037 MW at the 0.00108-MWh service floor. Experiment 24 independently completes 864 DC-SCED cells and 31,968 sample-contingency evaluations. Experiment 26 recomputes job-to-region and region-to-bus residuals and verifies the common witness digest before network settlement; its lineage certificate passes with zero typed residuals and explicitly labels the aggregate risk and payment profiles as analysis views.

VI. CONCLUSION

This paper couples a provenance-bound workload verifier with signed nodal settlement. On locked data, the joint profile reaches nRMSE/false credit/$F_1 = 0.323/1.786/0.453$ MWh/day, and its paired tail audit reduces the held-out false-credit ratio relative to the total-budget-only ablation (Fig. 3). The runtime-complete declaration witness covers all 75,326 submissions, conserves 1,972.115 MWh of declared runtime energy, shifts 41.1565 MWh from the event window, and carries one digest through 432 locked RTS-24 settlement cells and 37 finite N–1 outages per cell. The independent declaration replay confirms the response under a separate exact tariff policy. Fixed demand remains outside the flexible conversion, and deployment requires signed scheduler, telemetry, and topology attestation; field-causal validation remains a separate operational step. The aggregate risk fit is therefore reported as a statistical contract view, while the runtime-complete witness is the executable physical state used for network settlement.

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